

## # SER216 Final Project - Digital Parking Permit System

### Task 1: Requirements

#### Functional Requirements

1. The system shall allow users (students and staff) to register their vehicle details including license plate and model.
2. The system shall provide an interface for users to select and request parking permits for specific university zones.
3. The system shall enable security staff to verify permit validity by searching vehicle license plate numbers.
4. The system shall send an automated notification to the user once their permit request is approved or rejected by administration. #### Non-Functional Requirements

#### 1. Security:

1. Only authorized security personnel shall have access to update violation records.
2. Availability: The system shall be accessible 24/7 with a minimum uptime of 99.5% to ensure permits can be verified at any time. ---

### Task 2: Workflow Diagram (Description for diagrams.net)

\*Process Flow:\* [Vehicle Registration] -> [Permit Request] -> [Zone Check] -> [Permit Approval] -> [Parking Verification] ---

Task 3: Test Case Table | Test ID | Feature | Input / Data | Expected Result | Test Type | | :--- | :--- | :--- | :--- | | TC-01 | Vehicle Reg | Valid Plate & Model | Data saved to database successfully | System | | TC-02 | Zone Check | Selected Zone B | Display "Available" or "Full" status | System | | TC-03 | User Access | Student Login | User redirected to personal dashboard | Acceptance | | TC-04 | Permit Req | Submit Form | Admin receives "Pending" notification | Acceptance | | TC-05 | Profile Update | Change Phone Num | Permit remains valid after profile edit | Regression | | TC-06 | Security Search | Existing Plate | System displays "Permit Active" | Regression | --

### Task 4: Defect Analysis Defect:

"A valid parking permit is shown as expired during security verification." | Item | Student Answer | | :--- | :--- | |

#### Defect Category

| Logic / Database Error | |

#### Severity

| High (Prevents core security functionality) | |

#### Priority

| High (Needs immediate fix for daily operations) | |

#### Root Cause

| System comparing current date to a null or incorrectly formatted expiry field. | |

#### Regression Test

| Re-run TC-06 (Security Search) after applying the fix. | ---

### Task 5: GitHub Evidence \*Links and screenshots to be added here after execution.\*